Manuel Bonilla López, PhD

Engineering Manager & Technical Program Leader

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Personal Website | LinkedIn Profile | Google Scholar

EXECUTIVE PROFILE

Engineering manager and technical program leader with a Ph.D. in Physics and 16+ years of experience across defense and commercial semiconductor manufacturing, microelectronics, digital manufacturing, Manufacturing Execution Systems, optics, thin films, metrology, advanced materials, data systems, and research. Currently serve as an Associate Manager, Integrated Product Team Lead at L3Harris, leading 12 engineering technicians while carrying responsibility for people, cost, schedule, labor, resources, budgets, procurement, suppliers, risk, manufacturing readiness, and technical execution.

Maintain parallel hands-on leadership across Eyelit, the internally developed ZEUS platform, and Solumina/iBaseT. Current direct and advisory MES coverage spans five manufacturing locations, approximately 250-300 direct Eyelit and ZEUS users, and roughly 300 additional users enabled through trained site administrators and escalation support. Frequently brought into technically complex or stalled efforts where objectives, decision criteria, and ownership are unclear. Identifies the actual decision that needs to be made, translates complexity into clear but rigorous requirements, and creates the structure teams need to execute with confidence.

12	5	~600	~$150K / Year
Direct Reports	Supported Sites	Direct + Site-Enabled MES Users	Estimated Department Cost Avoidance

LEADERSHIP VALUE

- **People and organizational leadership:** Leads technical personnel, prioritizes workload and resources, develops capability, supports recognition and promotion cases, builds cross-training and backup ownership, and advocates for safe and sustainable work.
- **IPT and CAM program execution:** Integrates technical execution with EVMS-related cost and schedule ownership, labor forecasting, ETC/EAC inputs, annual budget planning, work authorization, procurement, suppliers, material needs, and program-risk communication.
- **Digital manufacturing and enterprise influence:** Connects headquarters, site leaders, administrators, vendors, developers, information technology, engineering, quality, and operations across MES implementation, governance, reporting, escalation, and adoption.
- **Structure from ambiguity:** Uses cross-disciplinary technical depth to identify the actual decision, separate essential requirements from unnecessary complexity, establish appropriate deliverables and decision criteria, and create an execution framework that enables teams to move forward confidently.

CORE CAPABILITIES

Management and Program Leadership	MES and Digital Manufacturing
People leadership | Engineering management | IPT leadership | CAM | EVMS | ETC/EAC | Cost and schedule management | Budget planning | Labor forecasting | Resource allocation | Operational risk | Executive communication	Eyelit MES | ZEUS | Solumina/iBaseT | Aegis Fusion | MES architecture | Digital travelers | Routes and workflows | Governance | UAT | Reporting | APIs | Equipment integration | Administrator enablement
Operations and Supplier Leadership	**Quality and Process Control**
Manufacturing operations | Procurement | Supplier qualification | Technical RFQs | Vendor evaluation | Precision manufacturing | 5-axis machining | CMM | Manufacturing readiness | WIP visibility | Cross-functional execution	PFMEA | SPC | OCAP | RCCA | Defect taxonomy | Yield | Change control | Risk assessment | Process health | Preventive maintenance | DOE | Qualification | Corrective action
Engineering & Advanced Manufacturing	**Data & Technical Tools**
Semiconductor manufacturing | ALD | MBE | E-beam evaporation | Sputtering | PECVD | CVD | Photolithography | Thin films | Optics | Vacuum systems | Tool ownership | Metrology	Python | SQL | R | Power BI | Data pipelines | Dashboards | Real-time monitoring | Spectrophotometry | Ellipsometry | STM | XPS/UPS | AFM/SEM

PROFESSIONAL EXPERIENCE

L3Harris — Palm Bay, Florida

– Associate Manager \| Integrated Product Team Lead — **Jun 2026 – Present**

- Lead 12 engineering technicians supporting platform preparation, cassette operations, and manufacturing processes; set priorities, allocate resources, coordinate workload, and balance operational needs against program cost and schedule commitments.
- Serve as the official IPT Lead and CAM for a manufacturing program workstream, integrating technical execution with budgeted labor, work authorization, cost and schedule visibility, resource forecasting, ETC/EAC inputs, procurement, suppliers, material needs, and program risk.
- Develop next-year budget requirements, monitor available and budgeted team hours, identify potential overruns or resource constraints, order required parts and materials, and coordinate actions with procurement and program leadership.
- Coordinate across Program Management, engineering, systems, manufacturing, quality, controls, tools, software, integration and test, procurement, operations, and organizations outside the direct reporting structure.
- Recovered a stalled supplier-qualification effort within two weeks by separating the vendor-facing questionnaire from the internal evaluation model, converting prescriptive tool and process questions into capability- and risk-based criteria, and distilling more than 60 proposed questions into a structured 22-question first-round screen; secured Program Manager approval without changes.
- Initiated a skills-matrix and cross-training structure to document current and prior capabilities, identify backup owners and single points of failure, improve staffing visibility, and create stronger evidence for employee development, internal mobility, recognition, and promotion decisions.
- Responded to an employee health concern involving extended soldering in an inadequately ventilated area by inspecting the work, pausing activity, engaging Environmental Health and Safety and management, and elevating broader gaps in ventilation, documentation, engineering assessment, personal protective equipment, and laboratory ownership.

– Concurrent Digital Transformation, MES and Manufacturing Reporting Leadership

Nov 2025 – Present

- Serve as an internal MES subject-matter resource and escalation-level consultant across Eyelit, the internally developed ZEUS platform, and Solumina/iBaseT, spanning implementation, governance, web configuration, reporting, data validation, training, adoption, troubleshooting, and manufacturing-workflow decisions.
- Support approximately 250-300 direct Eyelit and ZEUS users and roughly 300 additional users through site managers and trained administrators who resolve routine needs locally and escalate complex questions, architecture decisions, and unresolved issues.
- Drove Eyelit adoption and active use in Tempe, Mason, and Londonderry through executive demonstrations, technical and operational discussions, sandbox environments, local-administrator training, budget and scope planning, SOP and vendor implementation plans, roadmaps, role-based responsibilities, curricula, and approximately 40-slide training packages; continue to provide escalation-level consulting after enabling the sites to manage routine needs locally.
- Scaled the department implementation into an L3Harris corporate-supported MES solution, enabling multi-site Eyelit adoption while corporate completes a broader licensing model.
- Provide direct manufacturing-system support for approximately 50 State College users of ZEUS, including implementation workstreams, work instructions, MRB integration, team training, troubleshooting, developer and information-technology coordination, task ownership, schedules, and dependency tracking.
- Own development and remediation of a 17-report Solumina/iBaseT pilot backlog; when entering the effort, only two reports existed and both required correction. Translate manufacturing questions into requirements, data mappings, templates, validation logic, and operational reporting using JSreport, HTML, Handlebars, JSON, and MES API requests.
- Produced early report implementations with approximately 4-8-second execution in the observed test environment; one documented run retrieved 252 orders and displayed 183 active records in approximately eight seconds while supporting consolidated manufacturing visibility.
- Investigate APIs, pagination, status definitions, date and number formatting, field availability, quantity discrepancies, printing, PDF generation, shared links, filtering, sorting, and exports to determine whether system data supports the requested business metric.
- Contribute reporting and manufacturing-system expertise to the planned Solumina deployment across Palm Bay and Malabar; the future segment audience may reach thousands of users but is not represented as a current deployed-user population.
- Evaluated IPC-CFX and AMQP-based communication from automated optical-inspection equipment as an alternative integration path for sending manufacturing results into Solumina, including broker architecture, inspection data, image storage, and vendor-specific versus industry-standard interfaces.

– Semiconductor Senior Manufacturing Engineer

Mar 2024 – Nov 2025

- Assumed functional ownership of a stalled Eyelit MES implementation and brought the system from a nonfunctional state to production in approximately three months for roughly 200 users across operations, engineering, quality, metrology, and support functions.
- Architected digital routes, travelers, decision logic, data collection, Work in Process visibility, user roles, quality workflows, document control, SPC, defect tracking, reporting, UAT, training, governance, and sustainment; coordinated issues and enhancements through JIRA, vendors, developers, and information technology.
- Preserved the value of an estimated $300K software investment and approximately $200K in prior implementation effort while avoiding an immediate costly multiyear restart; the implementation's subsequent corporate sponsorship created approximately $150K per year in base-license cost avoidance for the local department.
- Led department-wide PFMEA across approximately 50 manufacturing tools and documented hundreds of process- and equipment-specific failure modes, effects, causes, prevention and detection controls, risk rankings, actions, owners, and completion evidence.
- Translated PFMEA findings into a defect taxonomy, failure classifications, severity, occurrence and detection criteria, and standardized resolution logic; defined the department OCAP connecting SPC triggers, tool shutdown, material disposition, engineering ownership, nonconformance escalation, RCCA, PFMEA updates, and closure.

- Designed and institutionalized the department's first end-to-end SPC and analytics framework, including control plans, standards, review cadence, metrics governance, dashboards, automated notifications, response expectations, and user training.
- Implemented environmental health monitoring using temperature and relative-humidity loggers, ISO-based limits, automated Python pipelines, live dashboards, and Microsoft Teams notifications.
- Built the facility workload model supporting approval of dedicated building support and helped create the annual preventive-maintenance calendar by reviewing manuals, interviewing tool owners, and converting required activities and frequencies into a visible schedule.
- Served on the Change Control Board and partnered with deposition engineering on e-beam chamber improvements, crystal-monitoring changes, process controls, equipment readiness, and safety risk assessments.
- Selected as a 2025 Space and Airborne Systems Operations Rising Star and received three L3Harris R.I.S.E. awards recognizing accelerated equipment risk-safety assessments, preventive-maintenance scheduling and real-time operational visibility, and Eyelit MES implementation, automated visualization, adoption, and sustainment.

Lockheed Martin — Orlando, Florida

Senior Optical Manufacturing Process Engineer — Jan 2022 – Jan 2024

- Served as the primary Group Lead for approximately four engineers and 10 engineering technicians whose functions included optical design, deposition, metrology, and cleaning; provided day-to-day technical and production leadership covering troubleshooting, work instructions, rework decisions, tool operation, parts planning, scheduling, and production support.
- Provided backup Group Lead coverage for a separate adjacent team of approximately five engineers and 10 engineering technicians, supporting technical decisions, work instructions, rework, tool operation, troubleshooting, and production planning without combining the two groups into a single direct-management claim.
- Served as coating-area subject-matter expert for Missiles and Fire Control and supported F-35, Sniper Advanced Targeting Pod, Javelin, Joint Air-to-Ground Missile, Modernized Day Sensor Assembly, Independent Research and Development initiatives, and other non-recurring programs.
- Led Independent Research and Development for diamond-like carbon and oxide coating systems and directed antireflective, high-reflective, and environmentally durable thin-film work across visible and infrared applications.
- Owned the lifecycle of customer-requested and non-recurring engineering work: optical modeling, budgeting, material selection and purchasing, equipment setup, recipe development, acceptance criteria, production execution, and shipment support.
- Designed coating processes to customer drawings, program specifications, MIL-PRF-13830 requirements where applicable, and internal production controls; owned Tecport Optics and Bühler Leybold PVD/CVD systems using e-beam evaporation, sputtering, thermal evaporation, and plasma-enhanced CVD.
- Diagnosed coating and equipment issues affecting yield, durability, and reliability, including delamination, non-uniformity, spectral deviation, environmental-test failures, chamber leaks, witness-sample absorption, and temperature-calibration errors.
- Applied structured RCCA to reduce defective coating runs from multiple failures per week to near zero within months and enabled delivery up to one month early on select programs.
- Recovered an environmental humidity test chamber after 3 months of downtime and a plasma-enhanced CVD system previously labeled to be out-of-production used for diamond-like carbon coatings; identified supplier and vendor quality gaps and led corrective actions aligning equipment, materials, and supplier capability with program requirements.
- Received four Next Gen Recognition Awards and the fourth-quarter 2022 Performance Management Team Advocate Award for collaboration, yield improvement, root-cause resolution, and improved operator-management issue resolution.

Intel Corporation — Hillsboro, Oregon

Semiconductor Process Engineer — **Jun 2020 – Sep 2021**

- Supported high-volume front-end semiconductor manufacturing at Intel's D1 fabrication facility as an atomic layer deposition tool owner, driving equipment reliability, film quality, process stability, troubleshooting, and technology-transition support.
- Owned two ALD manufacturing tools comprising four deposition sub-chambers for high-k dielectric film deposition supporting FinFET transistor production.
- Improved tool availability and process stability using design of experiments, SPC, chamber-pressure diagnostics, preventive-maintenance optimization, and pump-life analysis.
- Coordinated directly with equipment vendors to sustain high-volume production and resolve equipment-performance issues.
- Led conversion of two tools from a previous product line to the current production flow and restored inactive tools to production through coordinated diagnostics, vendor support, chamber recovery, and production qualification.

RESEARCH AND TECHNICAL EXPERIENCE

University of South Florida — Tampa, Florida

Research Associate — Advanced Materials Physics — **Aug 2013 – Jun 2020**

- Led the experimental work enabling the first reported room-temperature ferromagnetism in monolayer VSe_2, establishing a widely cited result in two-dimensional materials and magnetism.
- First author: "Strong Room-Temperature Ferromagnetism in VSe_2 Monolayers on van der Waals Substrates" Nature Nanotechnology 13, 289–293 (2018); independently highlighted in the Nature Nanotechnology editorial "2D Magnetism Gets Hot".
- Synthesized and characterized monolayer VSe_2 and developed deposition recipes for $TiSe_2$, $MoSe_2$, WSe_2, and $TaSe_2$ to study magnetic states, charge-density waves, excitonic behavior, band structure, doping, and surface morphology.
- Specialized in molecular beam epitaxy, physical vapor deposition, ultra-high-vacuum systems, scanning tunneling microscopy and spectroscopy, angle-resolved photoemission spectroscopy, vibrating-sample magnetometry, Raman spectroscopy, atomic-force microscopy, X-ray techniques, and electrical characterization.
- Installed, configured, maintained, and commissioned ultra-high-vacuum systems, including a shared departmental molecular beam epitaxy platform supporting multi-user research operations.
- Designed end-to-end experimental workflows and data-analysis systems and built predictive models linking process conditions to electronic, magnetic, structural, and surface behavior.

Guest-scientist appointments: Selected as guest scientist at SLAC National Accelerator Laboratory in California and SOLEIL Synchrotron in France for advanced characterization studies

University of Puerto Rico — Humacao, Puerto Rico

Research Associate — Materials Physics & Sensors — **Aug 2010 – Aug 2013**

- Conducted hands-on research in polymer electronics, nanofiber sensors, device fabrication, electrical characterization, and defense-related smart-textile applications.
- Fabricated gas-sensing diodes and transistors by electrospinning polymer nanofibers directly onto silicon wafers and depositing metal electrodes through thermal evaporation.
- Performed current-voltage and gas-response testing, used scanning electron microscopy to evaluate device morphology and sensing behavior, and developed polymer formulations and electrospinning recipes to improve device response.

University of Pennsylvania — Philadelphia, Pennsylvania

NSF REU Research Intern — Carbon Nanotubes & Graphene Devices — **Summer 2012**

- Grew carbon nanotubes and graphene using chemical vapor deposition and inspected samples to support nanoscale transistor-device fabrication.
- Used photolithography and metal deposition to fabricate devices and performed wafer-level electrical screening and current-voltage measurements to identify structures with carbon nanotubes or graphene bridging electrode gaps.
- Supported molecular-functionalization experiments, including applying DNA to selected devices and collecting before-and-after current-voltage measurements using established research procedures. Contributed experimental fabrication and characterization work to the peer-reviewed publication "Scalable, Non-Invasive Glucose Sensor Based on Boronic Acid Functionalized Carbon Nanotube Transistors."

University of Nebraska – Lincoln — Lincoln, Nebraska

NSF REU Research Intern — Surface Science & Electronic Materials — **Summer 2011**

- Prepared graphene–molecule and P3HT/P(VDF–TrFE) polymer-blend samples for experimental surface-science and electronic-materials research.
- Collected surface-sensitive spectroscopy measurements and cleaned, processed, and analyzed raw experimental data supporting investigations of molecular screening, electronic band offsets, band bending, and ferroelectric-poling effects.
- Co-authored two peer-reviewed publications— "Weak Screening of a Large Dipolar Molecule Adsorbed on Graphene" and "Changing Molecular Band Offsets in Polymer Blends of P3HT/P(VDF–TrFE) Due to Ferroelectric Poling"—and contributed to the American Physical Society presentation "Band Bending in Polymer Blends of PVDF-TrFE/P3HT by Poling the Ferroelectric Component."

AWARDS & RECOGNITION

- **L3Harris Rising Star Award (2025):** Selected as a Space and Airborne Systems Operations Rising Star, recognizing exceptional talent, initiative, rapid learning-to-action, proactive contribution, and tangible results within the first three years at L3Harris.
- **L3Harris R.I.S.E. Awards (3):** Recognized for accelerated equipment Risk Safety Assessments; preventive-maintenance scheduling and real-time operational visibility; and Eyelit MES implementation, automated visualization, adoption, and sustainment.
- **Lockheed Martin Next Gen Recognition Awards (4):** Recognized for cross-functional coating development, yield and measurement improvement, production and schedule performance, and root-cause/corrective-action execution.
- **Lockheed Martin Performance Management Team Advocate Award (Q4 2022):** Recognized for championing continuous improvement, strengthening operator engagement, and improving issue visibility and resolution.

EDUCATION & ACADEMIC RECOGNITION

- **Ph.D., Physics** | University of South Florida
- **M.Sc., Physics** | University of South Florida
- **B.Sc., Physics Applied to Electrical Engineering** | University of Puerto Rico
- Coursework concentration in Computational Mathematics | University of Puerto Rico
- Florida Louis Stokes Alliance for Minority Participation Graduate Fellowship
- American Physical Society Bridge Program Physics Graduate Fellow - Inaugural Cohort | University of South Florida

CERTIFICATIONS & CONTINUING EDUCATION

- **Integrated Product Team Lead (IPTL) Certification Program** | L3Harris Technologies
- **Google Data Analytics Professional Certificate** | Google via Coursera
- **Leading People and Teams Certificate** — **In Progress** | University of Michigan via Coursera
- **CS50 Computer Science Coursework — In Progress** | Harvard University via edX

TEACHING & TRAINING

- **Graduate Teaching Associate |** Physics Department, University of South Florida — Developed and delivered Physics I and II laboratory instruction, experimental setup, technical explanations, grading, and laboratory safety expectations.
- **Academic Tutor |** University of Puerto Rico — Supported undergraduate Calculus I–II, Precalculus I–II, and Physics I–II through university tutoring services.

RESEARCH IMPACT & EXTERNAL RECOGNITION

- **Publication record:** 10 peer-reviewed publications with over 2,500 scientific citations | Google Scholar
- **Career profiles:** American Physical Society News Bridge Program graduate profile (2020) | Symmetry Magazine PhD pathway feature (2015).
- **Research and community features**: University of South Florida feature on atomically thin magnetic semiconductors (2020) | El Oriental — "¡Orgullo Boricua!" doctoral profile (July 29, 2020, p. 12).